INCH-POUND

MIL-K-71129 (AR)
AMENDMENT 1
09 November 1994

MILITARY SPECIFICATION

KIT, REFURBISHMENT, FOR CARTRIDGE, 81MM
PRACTICE, SHORT RANGE, M880
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of MIL-C-71129, dated 26 May 1992, and is approved for use by the U.S. Army Armaments, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3: Delete: "propelling charge", and substitute: "ignition cartridge".

Delete 3.6.1 in its entirety and substitute:

"3.6.1 Misfires. The cartridges shall not fail to fire from the weapon when they are initially dropped into it."

Delete 3.6.4 in its entirety and substitute:

"3.6.4 Range and deflection precision. The cartridge shall have a horizontal range probable error (PEr) less than 15 meters, for all zones when calculated to the nearest one thousandth meter. The deflection probable error (PEd) shall be less than 15 meters for all zones when calculated to the nearest one thousandth meter."

PAGE 4

3.6.6: Add: "unobstructed", before the word "flight".

Add: "3.6.9 Dud plug function.

3.6.9.1 Functioned round. All dud plugs shall blow out of the projectile body upon fuze function.

3.6.9.2 Inert round. No dud plug shall exit the projectile body during ballistic flight or upon impact."

Add: "3.9 Dud plugs.

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"3.9.1 Configuration. Three dud plugs shall be present and properly inserted into the projectile body for the proving ground test.

3.9.2 Compliance. The dud plugs, prior to assembly to cartridges, shall comply with the criteria cited in paragraph 6.10."

PAGE 6

4.3.3: Delete "g" and "h" in their entirety and substitute:

"g. If the weapon is damaged by abnormal wear of the interior or of the firing pin, or caused by excessive pressure.

h. Two or more improper impact signatures occur."

Add the following:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

PAGE 7

Proving ground test line, under Conformance Criteria:

Delete: "72", and substitute: "96".

PAGE 9

4.4.1: Delete the word "interfix" from the text from lines a., b., c., and d.

Add the following:

"e. Dud plugs from one inspection lot that comply with 3.9.2."

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PAGE 11

4.4.2.1, Major 105: Change call out:

from: "Length of large outside diameter section",

to: "Length between large outside diameter aft face to firing pin point".

PAGE 16

4.4.2.6: Delete Major 103 in its entirety.

PAGE 20

4.4.2.10, under Special, add:

"G Any dud plug(s) missing (a) 3.9.1 Visual".

PAGE 21

4.4.3.1: Delete in its entirety and substitute:

"4.4.3.1 Proving ground test."

4.4.3.1.1, add:

"The refurbished cartridges shall be observed for the following characteristics:

a. Misfires, major defect (see 3.6.1)
b. Duds, major defect (see 3.6.5)
c. Short rounds, special defect (see 3.6.2)
d. Premature burst, critical defect (see 3.6.6)
e. Metal parts separation, special defect (see 3.6.3)
f. Precision, major defect (see 3.6.4)
g. Weapon damage, major defect (see 3.6.7)
h. Improper impact signature, major defect (see 3.6.8)
i. All dud plug heads out of any one projectile in the inert fuze ballistic test, Critical defect (see 3.6.9.2)
j. Two or more dud plug heads remain in the live fuze ballistic test, Major defect (see 3.6.9.1)".

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4.4.3.1.2, add:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

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Add the following:

"d. One group of 24 each with inert fuzes conditioned for at least 24 hours at 0 degrees F +/- 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3."

PAGE 25

6.10 Add the following:

"6.10 Dud plug useability criteria. The contractor shall provide dud plug material certification to the government for each inspection lot.

Each inspection lot of dud plugs shall be dimensionally inspected in accordance with the following chart.

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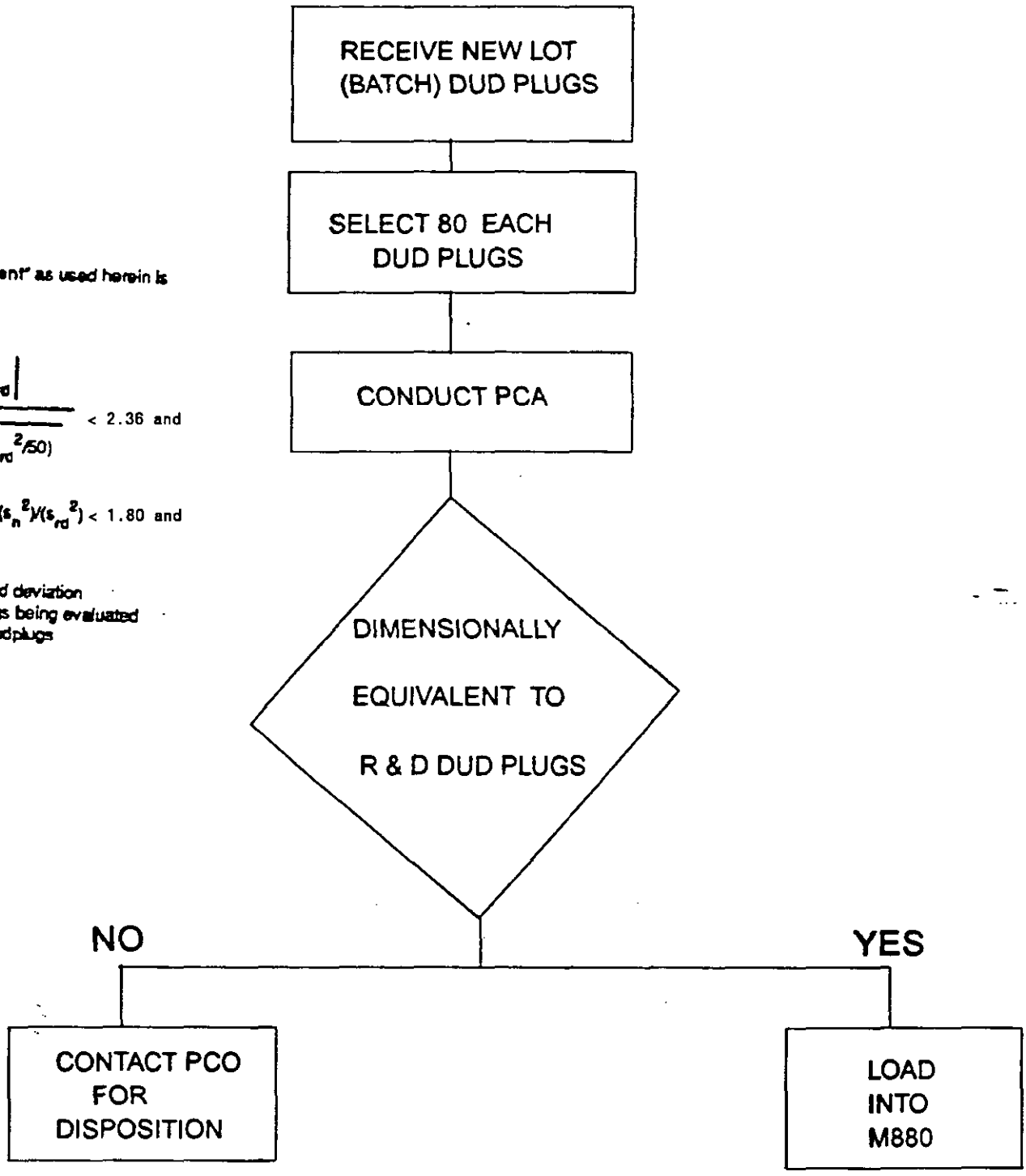
The term "equivalent" as used herein is defined as follows:

Dimensional:

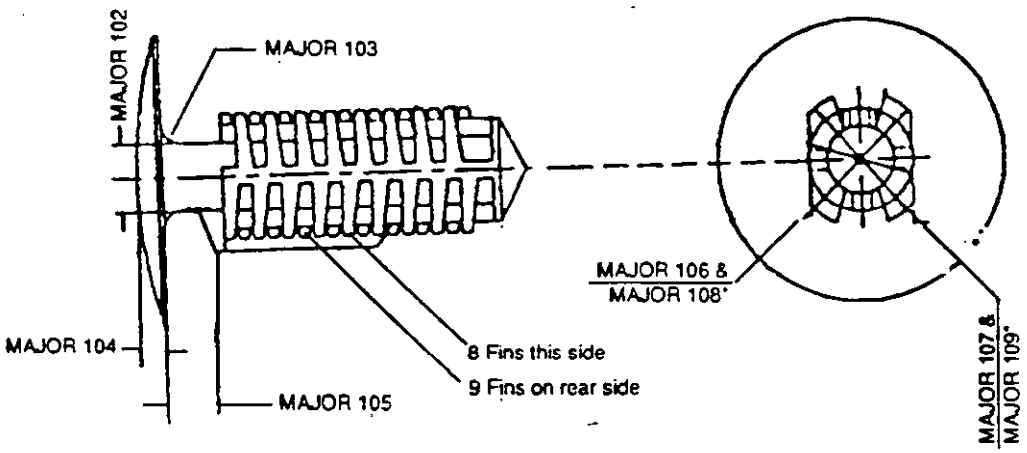
$$\frac{\left|\bar{X}_n - \bar{X}_{rd}\right|}{\sqrt{(s_n^2/80) + (s_{rd}^2/50)}} < 2.36 \text{ and}$$

$$(s_n^2)/(s_{rd}^2) < 1.80 \text{ and}$$

where: $\bar{X}$ = mean
s = standard deviation
n = dudplugs being evaluated
rd = R&D dudplugs



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* Dimensions are not symmetrical. In addition, the same dimension on the head end is smaller than the one on the first two fins at the point end. Majors 108 and 109 are the point end. Majors 106 and 107 are from the head end.

MEANS AND STANDARD DEVIATIONS TO USE FOR $\bar{X}_{rd}$ AND s_{rd} IN EQUIVALENCY FORMULAS ON SHEET ONE

DEFECT IDENTIFICATION	MEAN,in.	STANDARD DEVIATION,in.
MAJOR 102	.1470	.00073
MAJOR 104	.0585	.00100
MAJOR 105	.1365	.00077
MAJOR 106	.2883	.00092
MAJOR 107	.2869	.00088
MAJOR 108	.2967	.00090
MAJOR 109	.2975	.00124

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6.10.1 Dimensional inspection requirements. Examination for major and minor characteristics shall be performed in accordance with the dimensional inspections specified below.

Dud Plug (Drawing 12961001)

CLASS	CHARACTERISTIC	INSPECTION METHOD
MAJOR		
101	Plug cracked or deformed	Visual
102	Diameter of plug between head flat and fins (0.144 + .005)	SMTE
103	Radius at intersection of head and plug shaft	Visual
104	Distance from apex of head dome to head flat (.059 ± .010)	SMTE
105	Distance between head flat and first fin (.135 + .003)	SMTE
106	Larger cross sectional diameter of fins (head end) (0.288 ± .002)	SMTE
107	Smaller cross sectional diameter of fins (head end) (0.286 ± .004)	SMTE
108	Larger cross sectional diameter of fins (point end) (0.294 ± .005)	SMTE
109	Smaller cross sectional diameter of fins (point end) (0.295 ± .004)	SMTE
MINOR		
201	Poor workmanship	Visual"

6.11 Add the following:

"6.11 Inspection lot definition. A homogeneous collection of units of product or one shipment of component parts or assemblies from a subcontractor, bearing identification and treated as a unique entity from which a sample is to be drawn and inspected (and tested) to determine conformance with the acceptability criteria."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-0007)

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